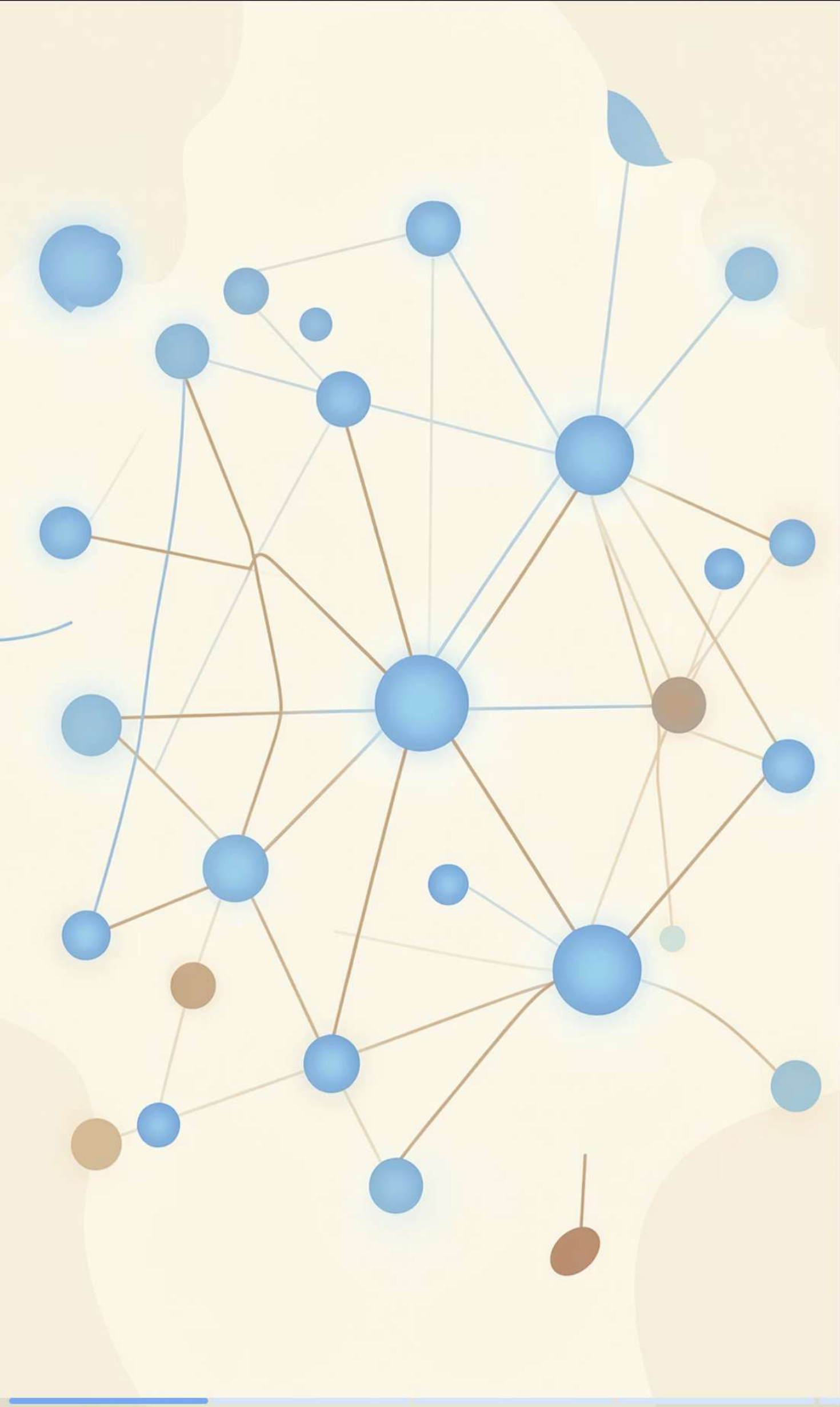




Evaluating the Impact of Feature Engineering on Machine Learning Model Performance

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BACKGROUND & MOTIVATION

Why Feature Engineering Matters



Focus on Data, Not Algorithms

Rather than chasing complex models, this project explores how thoughtful data transformation drives real performance gains.



Real-World Dataset

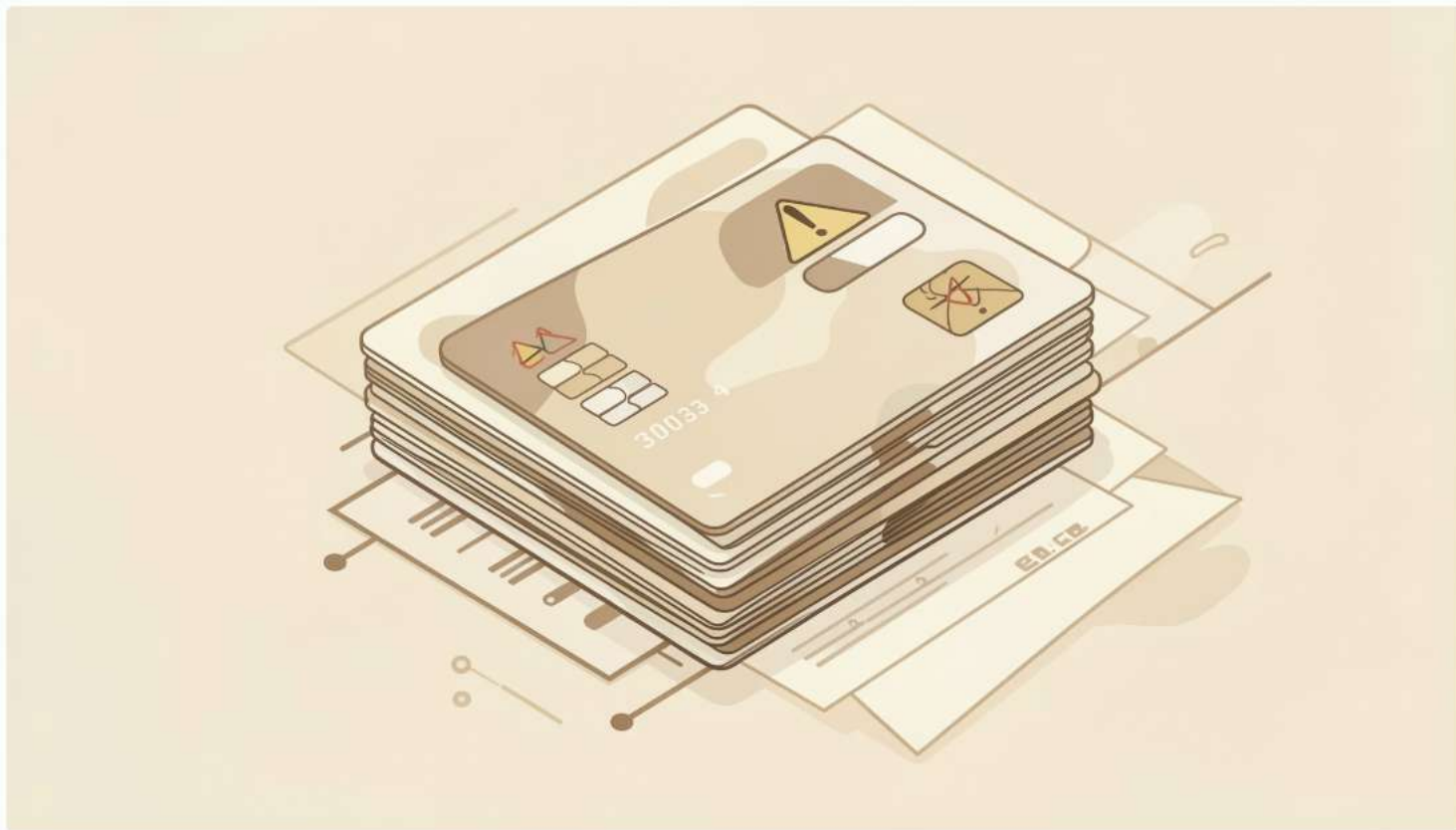
A widely-used UCI credit card default dataset provides a realistic, high-stakes testing ground for feature engineering strategies.



Clear Goal

Improve the detection of high-risk customers through better features - not just better algorithms.

Credit Card Default: A Costly Prediction Problem



Major Financial Risk

Credit card default is a leading source of loss for financial institutions - accurate prediction directly protects revenue.

Accuracy Is Not Enough

Models optimized purely for accuracy often miss the most critical cases - the actual defaulters.

Missing Defaulters Is Costly

False negatives - failing to flag a defaulter - translate directly into unrecovered debt and financial loss.

UCI Credit Card Default Dataset

30K

Customers

Real-world credit card holders

24

Features

Credit, payment, and demographic variables

6

Months

Sequential billing and payment history

What's Included

- Credit limit and loan amounts
- Monthly bill statements (6 months)
- Repayment amounts and history
- Demographic data: age, education, gender

Target Variable

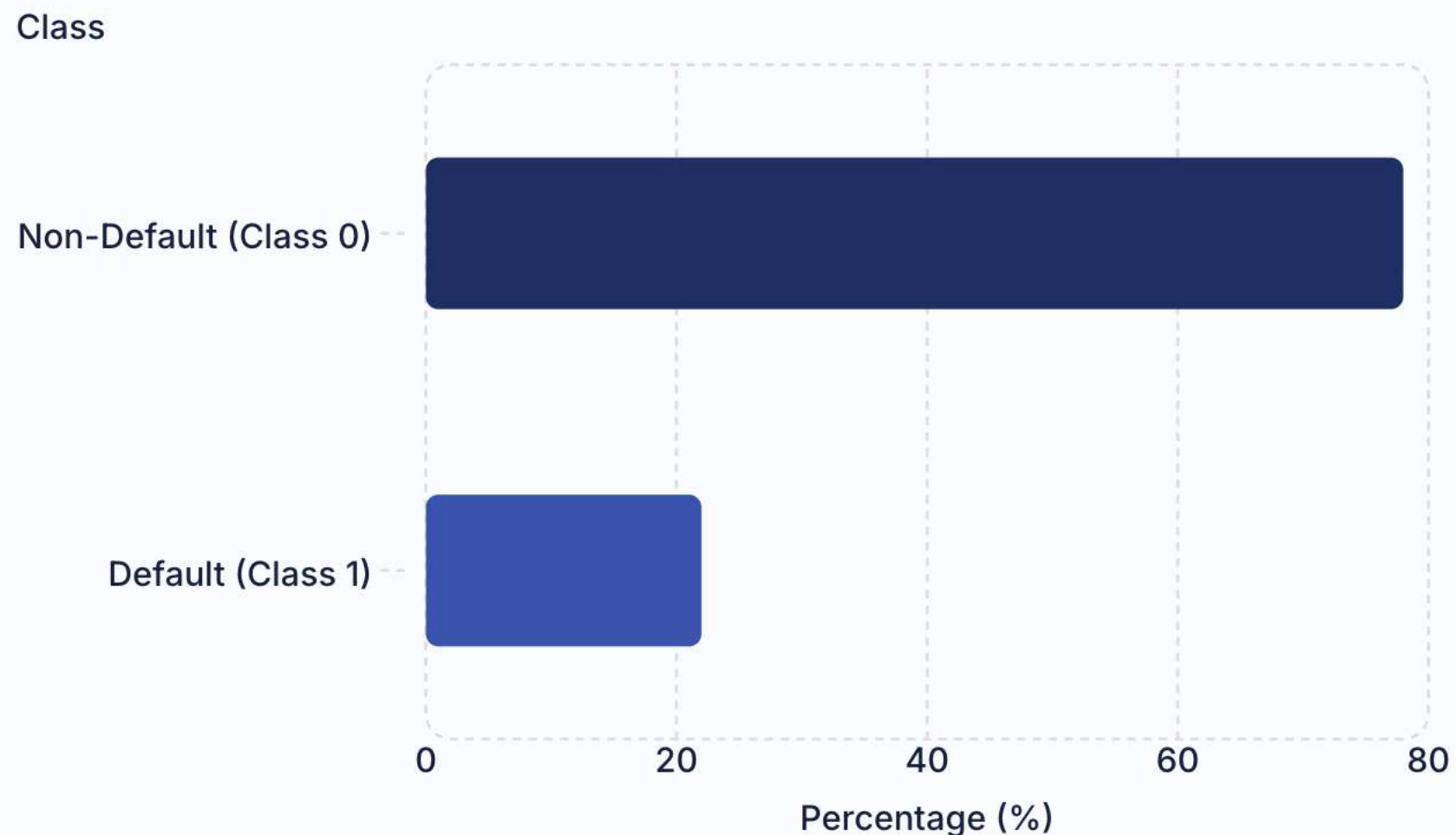


Default (Y)

0 = No default · 1 = Default

KEY CHALLENGE

The Class Imbalance Problem



Why This Matters

With **78% non-defaulters** and only **22% defaulters**, a naive model that always predicts "No Default" achieves 78% accuracy — while catching zero actual defaulters.

⚠ Accuracy is Misleading

High accuracy can hide poor detection of the minority class.

🎯 Recall is Critical

We must prioritize catching actual defaulters, even at some cost to precision.

Feature Engineering: Building Behavioral Signals



avg_bill

Average monthly bill amount -captures spending behavior over 6 months.



avg_payment

Average monthly payment - reflects repayment capacity and habits.



avg_delay

Average payment delay - a direct indicator of financial stress.

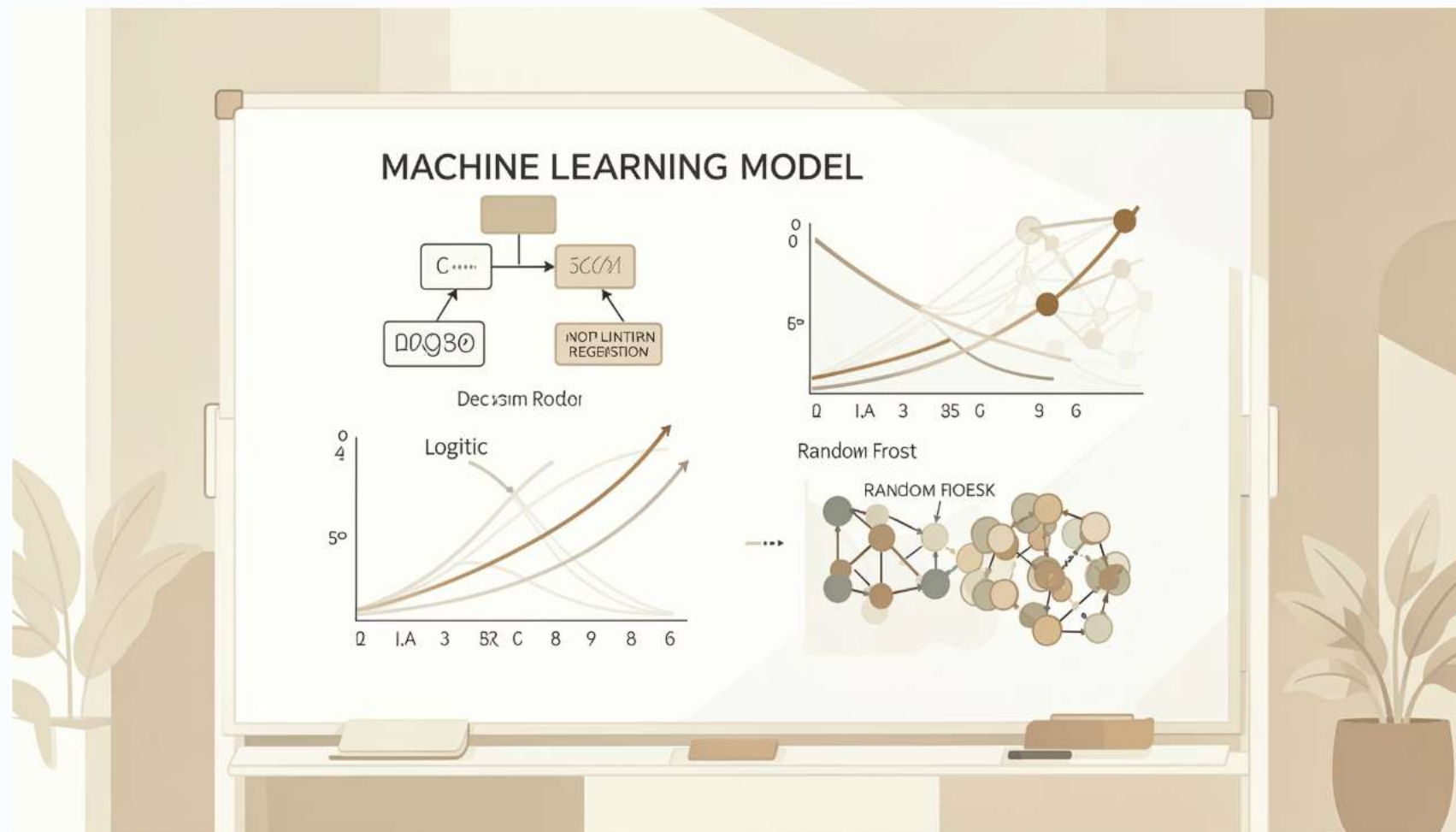


delay_count

Number of months with late payments - frequency of delinquency.

These engineered features reduce noise, improve interpretability, and encode meaningful financial behavior patterns that raw data alone does not capture.

Models Used: Simplicity & Interpretability



Logistic Regression

A transparent baseline model - easy to interpret and fast to train.

Random Forest

An ensemble approach that captures non-linear patterns without heavy tuning.

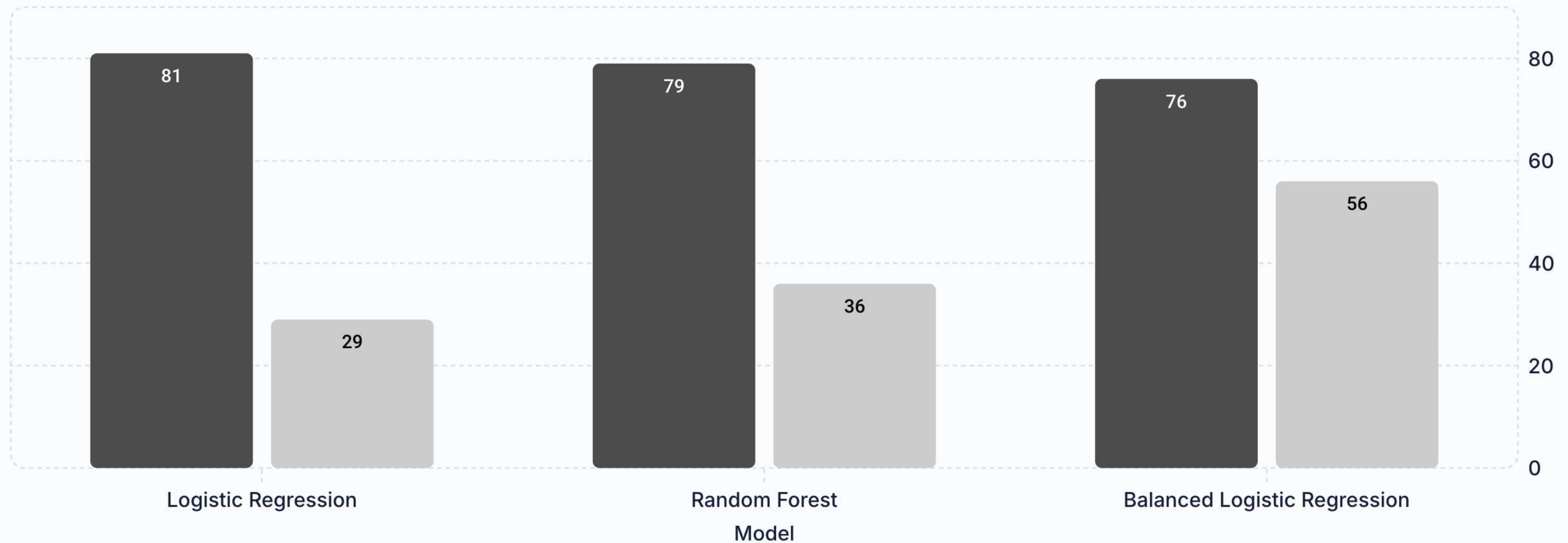
Balanced Logistic Regression

Logistic Regression with `class_weight` adjustment - explicitly penalizes missing defaulters.

- ☐ All models were trained on the same engineered feature set for a fair comparison.

Recall Comparison: A Significant Improvement

■ Accuracy (%) ■ Recall (%)



Accuracy Trade-off

Balanced LR drops from 81% → 76% accuracy - a small, acceptable cost.



Recall Gains

Recall nearly **doubles** from 29% → 56%, catching far more actual defaulters.



Best Model

Balanced Logistic Regression with engineered features delivers the strongest real-world performance.



KEY INSIGHTS

What the Results Tell Us

1

Feature Engineering Drives Performance

Behavioral features — not algorithm complexity — were the primary driver of improved recall.

2

Behavioral Signals Are Strong Predictors

Average delay and delay count emerged as the most powerful indicators of default risk.

3

Accuracy Alone Is Misleading

In credit risk, a high-accuracy model that misses defaulters is a failed model.

4

Recall Is the Right Metric

Prioritizing recall ensures the model serves its real business purpose: protecting against loss.

From Model to Real-World Application



Streamlit Web Application

A lightweight, interactive app enables real-time credit risk scoring — no coding required by end users.

Risk Classification

Instant "High Risk" or "Low Risk" output for each customer.

Probability Score

A calibrated probability gives decision-makers nuanced, actionable insight.

Identify High-Risk Customers

Proactively flag accounts before default occurs.

Reduce Financial Loss

Early detection enables intervention and loss mitigation.

Improve Decision-Making

Data-driven risk scores replace guesswork in credit decisions.

Future Enhancements & Roadmap



Model Improvements

- Use advanced models like XGBoost and LightGBM
- Perform hyperparameter tuning
- Explore ensemble techniques



Deployment & Scalability

- Deploy using cloud platforms (AWS/GCP)
- Enable real-time API integration
- Support large-scale processing



Explainability & Trust

- Add SHAP-based feature importance
- Provide clear reasoning for predictions
- Improve transparency for business users



User Experience

- Add batch predictions (CSV upload)
- Enable downloadable reports
- Build role-based dashboards



Data & Features

- Include transaction-level data
- Add time-based behavioral features
- Integrate external financial data sources



Risk & Compliance

- Address bias and fairness
- Ensure regulatory compliance
- Add audit logs for decisions